

**Project**

**Industrial Monitoring Controller**

**1. Purpose**

Define electrical requirements for the embedded monitoring controller.

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**2. Power Requirements**

| Parameter         | Value       |
|-------------------|-------------|
| Input Voltage     | 24V DC      |
| Power Consumption | <20W        |
| Backup Battery    | Lithium-Ion |
| Battery Life      | 24 Hours    |

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**3. Communication Interfaces**

- Ethernet
  - Wi-Fi
  - Bluetooth Low Energy
  - RS485
  - CAN Bus
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**4. Safety Requirements**

- Over-voltage protection
  - Reverse polarity protection
  - Short circuit protection
  - Surge protection
  - EMI shielding
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**5. Sensor Interfaces**

- Temperature
  - Pressure
  - Vibration
  - Humidity
  - Current Sensor
  - Voltage Sensor
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## **6. Compliance**

- IEC 61010
- CE
- RoHS
- FCC